

Department
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AI-BASED HEALTH RISK SCORING AND RESOURCE ALLOCATION

Authors: İrem Şahar, Abdullah Yusuf Erdem, Cemalettin Seyhan
Supervisor: Dr. Öğr. Üyesi Fatma Yertikaya Özkurt



MOTIVATION

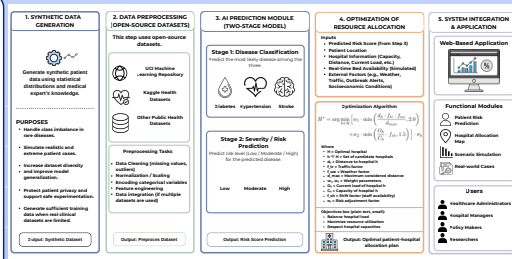
Noncommunicable diseases such as stroke, diabetes, and hypertension remain one of the leading causes of mortality worldwide. Many patients are diagnosed too late, while hospitals often face inefficient resource distribution and overload. Existing systems usually focus on either disease prediction or hospital management, but rarely combine both. Therefore, the main goal of this project is to develop an AI-based platform that combines early risk prediction with intelligent hospital routing to support faster and more efficient healthcare decision-making.

METHODOLOGY

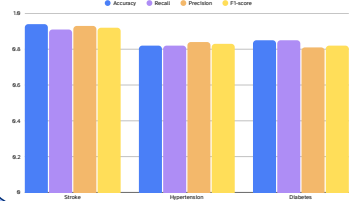
- Synthetic Data Generation and Data Preprocessing**
Generate synthetic patient data using statistical distributions and medical experts' knowledge.
- Exploratory Data Analysis (Open-source Datasets)**
- Two-stage ML Model Development**
ML model for predicting one of three diseases using synthetic data.
Diabetes: XGBoost classifier with probability calibration.
Hypertension: Logistic regression with quantile-based normalization.
Stroke: ANN (64-32-16) optimized for maximum recall.
- Real-World Validation**
Collection of real-world data via a 54-question online survey.
Statistically significant sample sizes determined using GPower (Power=0.95):
1,258 (Hypertension), **662** (Diabetes), **2,043** (Stroke) participants.
- Optimization Algorithm Simulation**
- System Integration and Evaluation**

OBJECTIVES

- Build** 2-stage machine learning pipeline based on non-laboratory data that can both identify the most probable disease and estimate individual risk levels.
- Design** an optimization-based patient routing algorithm that considers hospital capacity, distance, workload, and external factors within a simulated urban healthcare system.
- Reduce** waiting times, hospital overload, and misallocation of patients through improved routing and resource allocation strategies.
- Balance** the workload across hospitals, improving resource utilization and ensuring more equitable access to care.



RESULTS



LIMITATIONS

- Reliance on synthetic and limited open-source data may reduce real-world generalizability.
- Absence of real-time clinical data limits dynamic decision-making capabilities.
- Model performance may vary across populations and healthcare settings.
- Simplified simulation assumptions may not fully capture the complexity of real hospitals.

FUTURE ENHANCEMENTS

- Validate the framework using real-world clinical datasets.
- Incorporate real-time hospital and patient data streams.
- Extend the model to additional diseases and larger, diverse populations.
- Explore advanced models (e.g., deep learning, hybrid optimization methods).
- Deploy and test the system in real healthcare environments for practical evaluation.

OPERATIONAL WORKFLOW OF THE PLATFORM

